Submission

Solution

madrigal-protocol-lab is a local desktop laboratory for investigating an undocumented binary protocol over TCP: it reads a PCAP or PCAPNG capture, rebuilds each directional stream with per-byte provenance, applies declarative interpretation rules, verifies them on a corpus to surface counterexamples, and writes a portable project with a report.

Participant

Field	Value
Participant	Sabadash Pavel
Team size	one
Group	ИС-24

Hackathon

Field	Value
Hackathon	UMIRHack, 09-17 October 2026
Case	Protocol Laboratory, Madrigal

Setup

Python 3.12 or newer on Linux x86-64. Every command runs offline.

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
```

Run

Task	Command
Tests	`pytest tests/ -v`
Window	`python -m src.ui.main_window --capture tests/corpus/reference_export/corpus_capture_01.normalized.json --rule examples/corpus_rule_v1.json`
Pipeline	`python -m src.project.cli pipeline --pcap tests/corpus/corpus_capture_01.pcapng --project project.madrigal --report reports/REPORT.md`

Demo

The step-by-step window walkthrough is in `docs/demo.md`. The same investigation from the command line, used as the fallback, is in `docs/demo_cli.md`.

What is implemented

Area	Contents
`src/capture/`	PCAP and PCAPNG reading, session identification, directional reassembly with gap and ambiguity diagnostics, per-byte provenance, normalized export, CLI.
`src/hypothesis/` and `src/protocol/`	Framing strategies, declarative rules, the rule engine, corpus verification and counterexamples, the alternatives engine, rule versioning, CLI.
`src/ui/`	The PySide6 window: session tree, hex view with provenance, message comparison, rule and validation panels, the hypotheses panel, version diff.
`src/project/`	The portable on-disk investigation, relative paths and sha256, zip export and import.
`src/report/`	The investigation rendered to Markdown and to a self-contained HTML page.

Where things are

Path	Contents
`src/`	sources
`docs/`	reference, contracts, schemas, the audit and this file
`presentation/`	the defense deck and its screenshots
`tests/corpus/`	the research corpus and its reference export
`tests/fixtures/`	synthetic captures and deliberately damaged ones
`REPORT.md`	the investigation report

Limits

Area	State
IPv6	Supported; sessions, streams and checksums behave as for IPv4.
IP fragmentation	Reassembled for IPv4 and IPv6; an incomplete datagram is reported, never guessed.
Encrypted payloads	Out of scope; the bytes are reported as they appear.
Checksum offloading	Off by default; reordering is treated as a diagnostic.
Window project menu	Present; the Project menu creates, opens, saves, extends, exports, imports and verifies a portable project, and names and stores the current rule version, all delegating to `src.project`.
Organizer captures	Received and analysed; committed under `tests/corpus/real/`. Results are in `docs/REAL_CORPUS_ANALYSIS.md`.

Contact

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